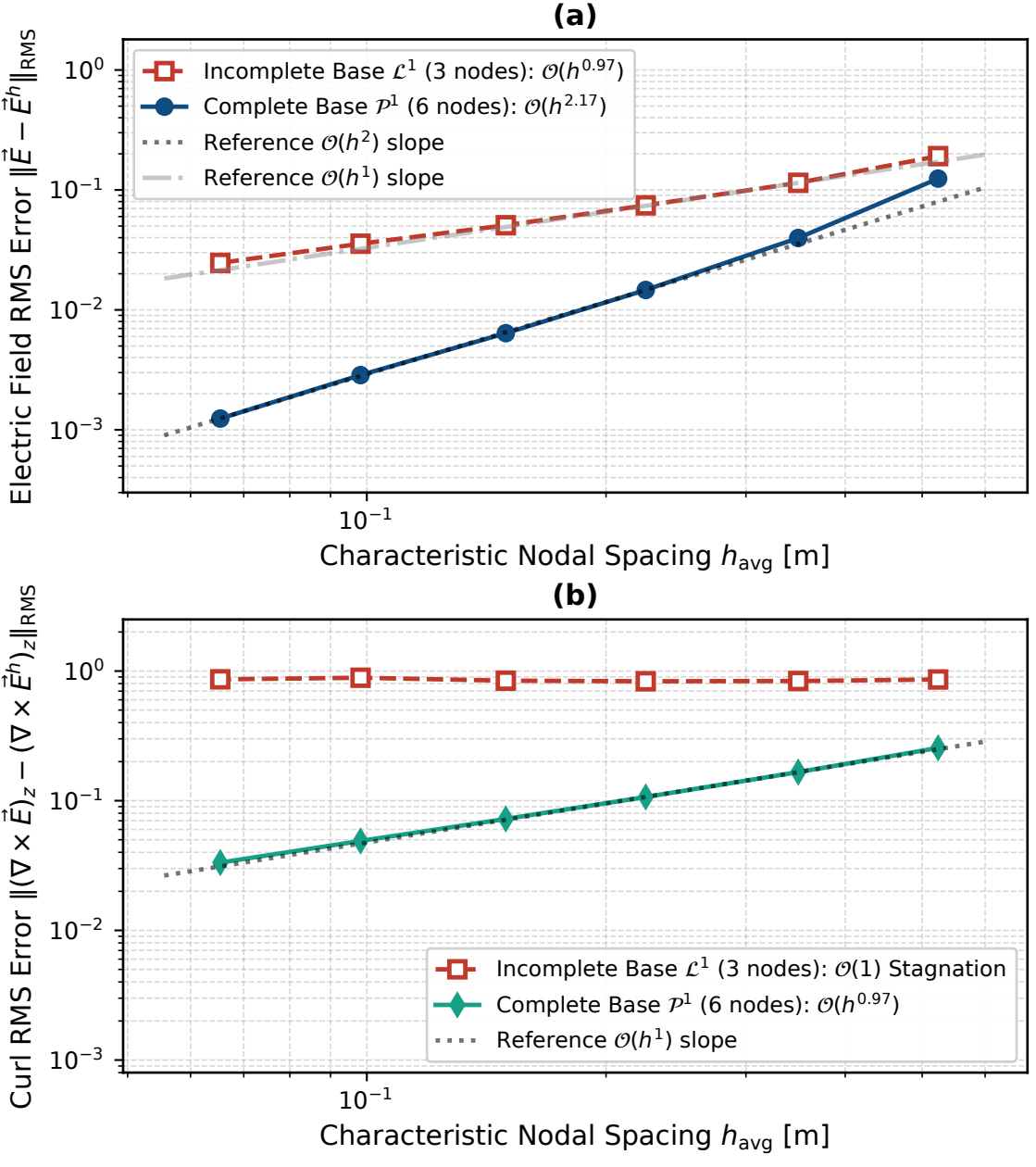
(a)
Incomplete Base $\mathcal{L}^1$ (3 nodes): $\mathcal{O}(h^{0.97})$
Complete Base $\mathcal{P}^1$ (6 nodes): $\mathcal{O}(h^{2.17})$
Reference $\mathcal{O}(h^2)$ slope
Reference $\mathcal{O}(h^1)$ slope
Electric Field RMS Error $\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$
$10^0$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-1}$
Characteristic Nodal Spacing $h_{\text{avg}}$ [m]
(b)
Incomplete Base $\mathcal{L}^1$ (3 nodes): $\mathcal{O}(1)$ Stagnation
Complete Base $\mathcal{P}^1$ (6 nodes): $\mathcal{O}(h^{0.97})$
Reference $\mathcal{O}(h^1)$ slope
Curl RMS Error $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\text{RMS}}$
$10^0$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-1}$
Characteristic Nodal Spacing $h_{\text{avg}}$ [m]